

ALI AL SALEM AB, Kuwait

WMO: 405736

Lat: 29.347N

Lon: 47.521E

Elev: 472

StdP: 14.45

Time Zone: 3.00 (E03)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.4	42.5	10.5	9.6	66.1	14.0	11.4	71.3	25.7	61.7	23.1	61.6	8.2	280	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.5	120.0	70.5	118.0	69.5	115.3	68.7	80.6	98.6	77.0	98.9	73.5	99.4	14.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	135.2	93.9	70.5	114.1	90.0	65.9	97.1	84.5	44.7	100.5	40.7	101.0	37.3	97.0	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	23.6	21.0	34.6	122.2	2.3	2.1	33.0	123.7	31.7	125.0	30.4	126.1	28.7	127.7
			29.4	82.0	3.9	3.0	26.6	84.2	24.3	86.0	22.1	87.7	19.2	89.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	80.4	56.0	60.0	68.4	80.2	91.5	98.6	101.4	101.3	94.5	84.8	68.0	58.9	(6)
(7)		DBStd	17.52	5.40	6.05	6.77	6.33	5.76	3.88	3.51	3.05	4.26	5.63	7.12	5.81	(7)
(8)		HDD50	19	10	4	0	0	0	0	0	0	0	0	0	5	(8)
(9)		HDD65	725	281	157	39	1	0	0	0	0	0	0	47	200	(9)
(10)		CDD50	11123	196	285	570	905	1288	1459	1595	1590	1336	1080	539	280	(10)
(11)		CDD65	6355	3	18	144	456	823	1009	1130	1125	886	615	136	10	(11)
(12)		CDH74	105375	20	139	1436	5589	13375	18476	21078	20708	15099	8213	1161	81	(12)
(13)		CDH80	76109	2	31	540	3002	9246	14174	16617	16246	10918	4920	408	5	(13)
(14)	Wind	WSAvg	10.5	9.1	10.0	10.3	10.1	10.6	13.3	13.5	11.4	10.4	9.1	8.8	8.8	(14)
(15)	Precipitation	PrecAvg	4.4	0.9	0.4	0.6	0.4	0.1	0.0	0.0	0.0	0.0	0.1	0.7	0.9	(15)
(16)		PrecMax	8.2	3.7	2.7	2.5	1.8	1.0	0.0	0.0	0.0	0.4	1.5	4.5	2.7	(16)
(17)		PrecMin	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.8	0.9	0.6	0.7	0.4	0.2	0.0	0.0	0.0	0.1	0.3	1.1	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.7	84.2	98.9	107.2	118.1	121.8	123.7	122.1	116.8	107.8	94.6	79.7	(19)
MCWB			56.4	57.0	59.7	69.8	76.2	75.6	71.5	71.1	69.9	67.0	62.7	56.3	(20)	
2%		DB	71.8	77.3	91.2	100.7	112.9	118.1	120.1	118.7	113.3	104.2	88.2	76.7	(21)	
		MCWB	54.3	56.4	58.9	66.6	70.2	69.4	69.6	70.0	67.6	65.1	62.5	55.9	(22)	
5%		DB	69.4	74.8	85.7	97.1	109.4	115.1	118.0	116.9	111.1	102.0	84.4	73.3	(23)	
		MCWB	54.2	55.1	58.3	64.0	67.3	68.3	68.7	69.4	67.0	64.5	59.9	56.0	(24)	
10%		DB	66.2	71.5	80.9	93.5	105.9	112.9	115.1	115.0	108.0	98.8	80.7	69.9	(25)	
		MCWB	53.4	53.9	57.0	62.8	66.3	67.5	68.4	68.7	66.1	63.9	59.2	54.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.6	63.7	66.1	74.3	82.8	84.1	81.1	84.1	80.7	77.8	72.8	64.8	(27)
MCD B			66.9	68.8	76.3	98.9	114.4	117.2	98.6	100.1	91.4	86.0	78.7	69.3	(28)	
2%		WB	60.5	61.4	64.0	69.7	75.5	76.2	76.5	80.9	73.2	74.9	69.1	62.2	(29)	
		MCD B	65.2	69.2	76.0	90.1	102.1	109.8	103.5	95.6	94.9	83.0	77.9	68.2	(30)	
5%		WB	58.0	59.5	62.0	67.5	70.9	70.3	72.2	77.2	69.5	71.7	65.7	59.8	(31)	
		MCD B	63.9	68.2	76.2	86.8	96.1	108.6	111.1	101.2	106.2	83.9	74.8	66.4	(32)	
10%		WB	55.5	57.4	59.7	65.8	68.3	67.7	69.6	72.3	67.7	68.7	63.3	57.4	(33)	
		MCD B	63.0	65.9	77.7	85.8	96.3	111.2	113.1	109.1	104.6	86.2	73.8	66.6	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.7	19.9	23.1	24.2	25.7	27.6	27.5	27.9	27.0	24.9	20.4	19.9	(35)
MCD BR			21.4	23.1	28.7	28.6	31.8	30.5	30.6	30.0	28.8	28.6	24.0	22.9	(36)	
MCWBR			10.2	9.8	11.1	11.0	12.6	11.5	10.6	10.5	10.2	11.3	10.7	11.9	(37)	
5% WB		MCD BR	16.7	18.1	21.2	24.6	29.1	31.6	30.7	27.9	28.0	22.3	18.5	16.7	(38)	
		MCWBR	10.3	10.1	10.5	10.7	15.0	15.3	13.7	14.7	11.7	13.0	10.7	10.2	(39)	
																(39)
(40)	Clear-Sky Solar Irradiance	taub	0.495	0.543	0.600	0.690	0.695	0.648	0.662	0.613	0.563	0.567	0.537	0.498	(40)	
taud		1.786	1.689	1.573	1.467	1.466	1.514	1.514	1.600	1.686	1.683	1.732	1.792	(41)		
Ebn at Noon		223	225	223	210	210	219	215	224	230	215	208	214	(42)		
Edh at Noon		59	70	84	97	97	92	92	83	74	70	62	56	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1126	1404	1767	1931	2221	2488	2393	2273	2017	1593	1172	1039	(44)	
(45)		RadStd	80	84	70	95	98	83	64	48	64	68	92	91	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (11 neighbors)	+0.70	N/A	N/A	+1.06	N/A	N/A	N/A	N/A	+232	+245	(47)

Nomenclature: See separate page